Electrical simulation: findings and fixes

A reproducible SPICE experiment for the proposed extension. The existing Phase 3 circuit was not modified or energized.

SIMULATION CANDIDATE - bounded models only; no construction, procurement or physical commissioning release

Result within the declared model envelope

48 required scenarios pass; 6 unsafe comparison cases detected.
Each accepted transient reaches its requested end time.
R/L/diode/switch equations plus behavioral protection ICs.

What the simulation does not establish

No selected-component SOA, thermal or contact certification.
No fluid dynamics, actual pump control electronics or valve motion.
No physical pressure target, wiring sign-off or final priced basket.

Rejected comparison circuits and resulting candidate changes

Insufficient transmitter supply

12 V + 150 ohm cannot maintain the requested 20 mA.
24 V +/-10% instrument loop; 100 ohm burden.

ADC overrange and power-off backfeed

24 mA / 150 ohm exceeds the normal 3.3 V input rail.
Current limit + 10 kohm + clamp + power-off isolation.

Unclamped inductive turn-off

Negative control reaches assumed avalanche at 100.2 V at a drain.
Opposed diode/avalanche clamp on each modeled load.

Automatic restart after E-stop

Deliberately unsafe reset logic is rejected.
Fresh reset edge with permit low; heartbeat and power-good.

Combined master/contact and transistor faults

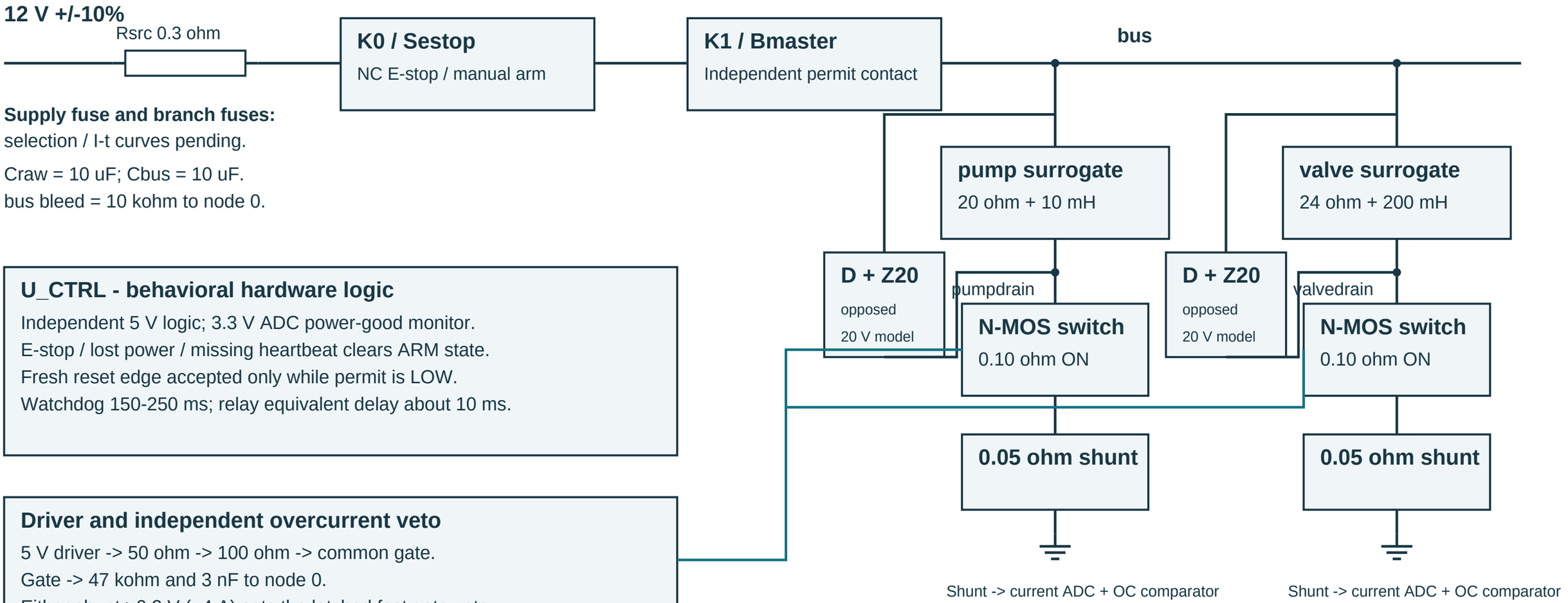
The combined welded master + shorted MOSFET remains unsafe.
Retain independent E-stop cutoff; no claim of universal fault tolerance.

Negative controls are intentional experiments, not reconstructions of the existing Phase 3 bench.

R1 candidate power-stage schematic

Node names and component values map to the included power netlists; U_CTRL and IC behavior are functional equivalents.

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Supply fuse and branch fuses:
selection / I-t curves pending.

Craw = 10 uF; Cbus = 10 uF.
bus bleed = 10 kohm to node 0.

U_CTRL - behavioral hardware logic

Independent 5 V logic; 3.3 V ADC power-good monitor.
E-stop / lost power / missing heartbeat clears ARM state.
Fresh reset edge accepted only while permit is LOW.
Watchdog 150-250 ms; relay equivalent delay about 10 ms.

Driver and independent overcurrent veto

5 V driver -> 50 ohm -> 100 ohm -> common gate.
Gate -> 47 kohm and 3 nF to node 0.
Either shunt >0.2 V (~4 A) sets the latched fast gate veto.

Design requirements exposed by this model

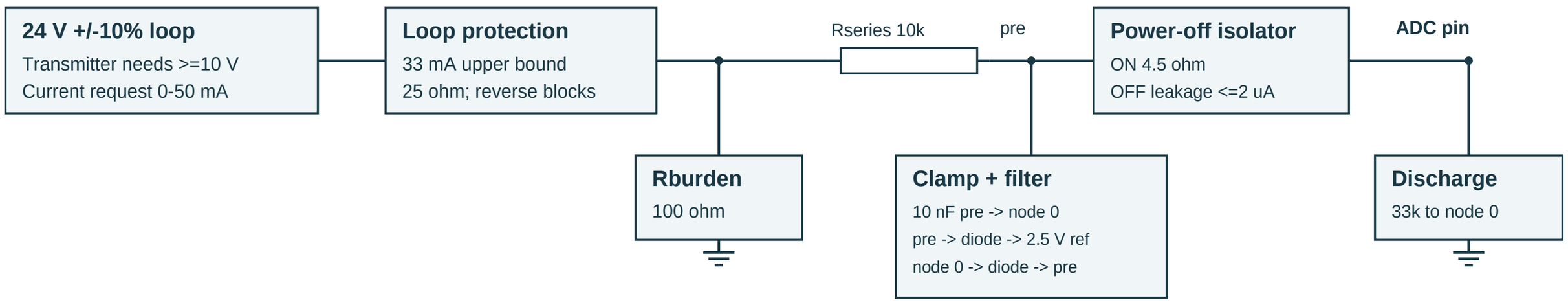
K0 and K1 are separate physical interruption paths; a welded K1 plus shorted MOSFET is an uncovered combined fault.
100 V generic avalanche envelope / 70 V design criterion is not a selected MOSFET rating or a verified safe operating area.
Low-side shunts measure switch current, including startup; they do not measure the full recirculating flyback tail.
Source is a Thevenin supply. Real SMPS foldback, fuses, coil-driver transistors, GPIO translation and wiring parasitics remain unmodeled.
No motor or valve power may be routed through the preserved Phase 3 indicator/relay circuit.

R1 candidate sensor interfaces

Protection topology is electrically simulated. Named commercial devices are candidates, not transistor-level macromodels.

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Pressure current-loop receiver



Reference / ADC equivalent

5 V conditioner -> 1k -> 2.5 V shunt reference.
ADC: 6 Mohm input, 10 pF and supply ESD diodes.
Isolator opens if 3.3 V <3.0 V or conditioner <4.5 V.

Calibration consequence

10k/33k network loads the burden; use end-to-end calibration.
Modeled 4 mA -> 0.305839 V; 20 mA -> 1.529191 V.
Values are simulated electrical gain, not physical pressure calibration.

Pulse flow / isolated bus-voltage observation

Field input

Flow: 4.5-5.5 V -> 560 ohm
Bus: 12 V -> 1.8 kohm
LED + reverse diode

Bounded optocoupler

CTR envelope 0.2 to 2.0
3.3 V pullup 10k / 1 nF
Model saturation is continuous

3.3 V logic input / source identity

Active-low pulses; 500 Hz waveform simulated.
No pulses can mean zero flow OR a broken signal wire.
Voltage feedback is supply presence, not valve position.

Normal 4-20 mA readings must remain within the ADC supply rails; absolute-limit survival alone is not accuracy.

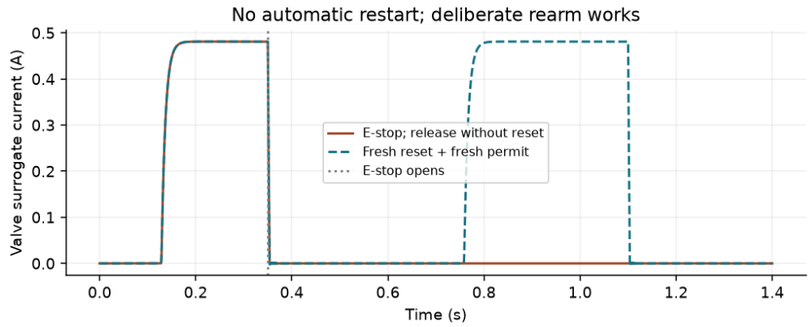
Reverse polarity / open loop / short burden are not classified as valid process measurements.

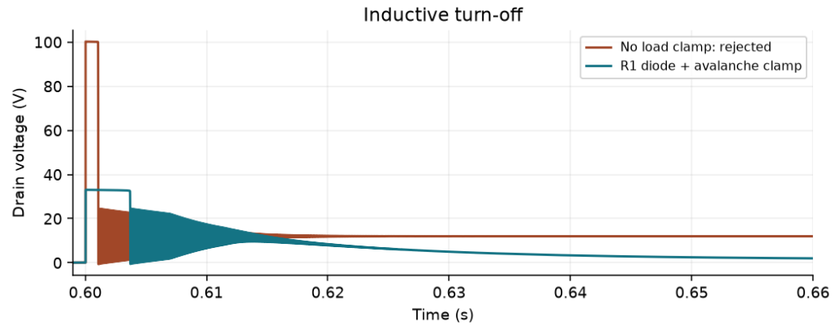
Temperature / I2C wiring, board pullups and GPIO driver backfeed are outside this simulation pack.

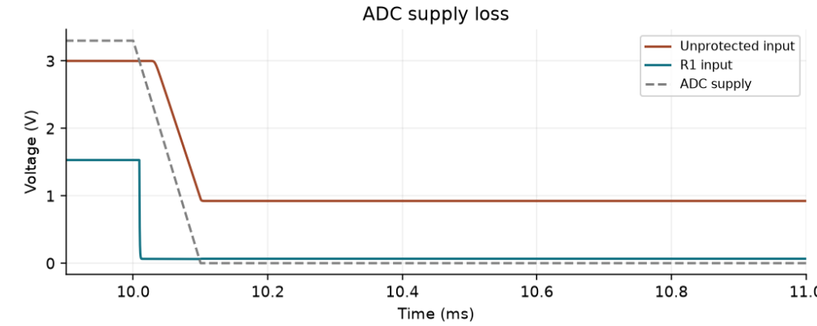
Verification evidence and remaining release gates

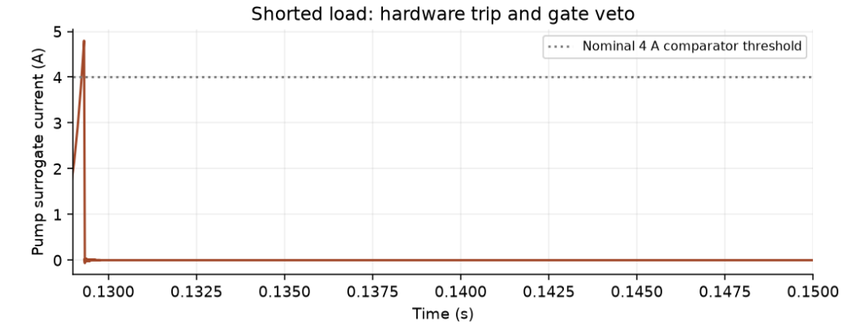
Each case includes its exact netlist, simulator log, compressed sampled trace and machine-readable acceptance checks.

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Measured in the model

Normal case peak drain 33.06 V; shorted-pump case peak current 4.80 A.

ADC-off case final input 0.0656 V; post-settling rail injection remains below 1 uA.

The 1 us rail-fall case also meets the modeled absolute input limits; brief injection is separately retained in its trace.

No automatic restart after E-stop, brownout or logic loss; fresh reset with permit low restores a controlled run.

20 us repeats cover E-stop and shorted load; the short repeat uses first-order integration. This is a numerical check.

Before selecting and building actual hardware

1. Freeze pump/valve datasheets, real coil/inrush/BLDC behavior, MOSFET SOA, suppression and fuse coordination.
2. Realize and independently review the watchdog, reset latch, coil drivers, GPIO interfaces and protected ADC circuit.
3. Review bench photographs and continuity; issue the actual pin/terminal wire schedule and matched <=INR15,000 basket.
4. Perform unpowered checks, dry electrical tests, passive leak checks, read-only calibration and attended protected trials.

Primary source constraints: TI ADS1115 / TMUX1511; ADI MAX14626. Model inventory and source links: README.md.